Solute transport

SWAT+ Documentation

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Purpose

The rtb-salt module simulates the eight salt ions (so4, ca, mg, na, k, cl, co3, hco3) across the soil profile, plant, channel, aquifer, urban, irrigation, fertilizer, atmospheric, and road-salt pathways. Each input file listed below feeds one piece of that state: initial concentrations, applied loads, or per-plant tolerance parameters.

Activation

Salt simulation is gated by the master constituent file `constituents.cs` referenced from `file.cio` (`in_sim%cs_db`, default `constituents.cs`). The number of salt ions in that file is held in `cs_db%num_salts`. If `num_salts = 0`, salt readers that test it (`salt_roadsalt_read`, `salt_uptake_read`, `salt_urban_read`) skip silently. The remaining readers test only for file existence and read whatever is present.

All readers are called from `proc_read.f90` except `salt_cha_read`, which is called later from `main.f90`.

Files

salt_hru.ini

Initial salt ion concentrations and mineral fractions for HRU soils and plants.

- Reader: `src/salt_hru_read.f90`
- Type: `salt_soil_ini` (`type cs_soil_init_concentrations`) in `src/constituent_mass_module.f90`
- Format: 4 header lines, then for each entry three lines: name, then `num_salts + 5` soil values, then `num_salts + 5` plant values. First `num_salts` values are ion concentrations (ppm), next 5 are salt mineral fractions.

salt_aqu.ini

Initial salt ion concentrations and mineral fractions for aquifers.

- Reader: `src/salt_aqu_read.f90`
- Type: `salt_aqu_ini` (`type salt_aqu_init_concentrations`) in `src/constituent_mass_module.f90`. Fields: `name`, `conc(num_salts)` in g/m3, `frac(5)` mineral fractions.
- Format: 4 header lines, then one line per aquifer init record: `name`, `num_salts` concentrations, 5 mineral fractions.

salt_channel.ini

Initial salt ion concentrations for channels.

- Reader: src/salt_cha_read.f90
- Type: `salt_cha_ini` (`type salt_cha_init_concentrations`) in `src/constituent_mass_module.f90`. Fields: `name`, `conc(num_salts)` in g/m3.
- Format: 2 header lines, then one line per record: `name`, `num_salts` concentrations.

salt_irrigation

Salt ion concentrations for irrigation water sourced from outside the watershed.

- Reader: src/salt_irr_read.f90
- Type: `salt_water_irr` (`type cs_irrigation_concentrations`) in `src/constituent_mass_module.f90`. Fields: `name`, `water(num_salts)` in ppm.
- Format: 2 header lines, then one line per source: `name`, `num_salts` concentrations.

salt_plants

Plant salt-tolerance parameters and stress control switches.

- Reader: src/salt_plant_read.f90
- Targets in src/salt_data_module.f90:

Field	Type	Units	Description
salt_tds_ec	real	none	TDS to electrical conductivity conversion factor
salt_tol_sim	int	flag	0 = off, 1 = simulate salt effect on plant growth
salt_soil_type	int	code	1 = CaSO4 soils, 2 = NaCl soils
salt_effect	int	code	1 = applied after other stresses, 2 = combined with other stresses via min
salt_stress_a(plantparm)	real	none	a parameter in the salinity relative-yield equation
salt_stress_b(plantparm)	real	none	b parameter in the salinity relative-yield equation

Format: title line, header, `salt_tds_ec`, blank, `salt_tol_sim`, `salt_soil_type`, `salt_effect`, four skipped lines, header, then one line per plant in plant database order with `name, a, b`.

salt_road

Road salt loadings (kg/ha) per atmospheric deposition station.

- Reader: src/salt_roadsalt_read.f90
- Target: `rdapp_salt(:)%salt(:)` (`%road`, `%roadmo`, `%roadyr`, `%roadday`). Allocated for `0:atmodep_cont%num_sta` stations.
- Format: 4 commentary lines skipped, then per station and per salt ion either an annual value (`aa`), a monthly series (`mo`), or a yearly series (`yr`). For `yr`, the reader internally splits the annual mass to daily loadings on days with precipitation and sub-zero temperature.

salt_uptake

Specified daily salt uptake (kg/ha) per plant community per ion.

- Reader: src/salt_uptake_read.f90
- Targets in `src/salt_module.f90`: `salt_uptake_kg(plantparm, num_salts)`, `salt_uptake_on` (flag set to 1 when the file is present).
- Format: 3 header lines, then one line per plant: `name`, `num_salts` uptake values.

salt_urban

Salt ion concentration in suspended solids from urban impervious areas (mg salt / kg sediment).

- Reader: src/salt_urban_read.f90
- Target: `salt_urban_conc(urban, num_salts)` in `src/salt_module.f90`.
- Format: 2 header lines, then one line per urban land-use type: `urb_type`, `num_salts` concentrations. The urban name must match an entry in `urban.urb`.

salt_fertilizer.frt

Per-fertilizer salt ion loading (kg/ha).

- Reader: src/salt_fert_read.f90
- Type: `fert_salt(fertparm)` (`type fert_db_salt`) in `src/salt_module.f90`. Sets `fert_salt_flag = 1`.

#	Field	Units	Description
1	fertnm	-	fertilizer name
2	so4	kg/ha	sulfate loading
3	ca	kg/ha	calcium loading
4	mg	kg/ha	magnesium loading
5	na	kg/ha	sodium loading
6	k	kg/ha	potassium loading
7	cl	kg/ha	chloride loading
8	co3	kg/ha	carbonate loading
9	hco3	kg/ha	bicarbonate loading

Format: title, header, then one line per fertilizer in the fertilizer database order.

Related

- constituents.cs. Master constituent list. Sets `num_salts` and the ion names.
- fertilizer.frt. The base fertilizer database. `salt_fertilizer.frt` must have the same row count.
- urban.urb. Source of urban land-use names referenced by `salt_urban`.
- atmodep.cli. Defines the atmospheric deposition stations used by `salt_road`.

Purpose

The rtb-cs module simulates user-listed water-quality constituents (selenium SeO4 and SeO3, boron, and additional species defined in the constituent database). It runs in parallel with rtb-salt and shares the same master file, `constituents.cs`.

Activation

`constituents.cs` is the master file referenced from `file.cio` as `in_sim%cs_db` (default `constituents.cs`). It lists the pesticides, pathogens, metals, salt ions, and other constituents in the simulation, and populates `cs_db%num_pests`, `cs_db%num_paths`, `cs_db%num_metals`, `cs_db%num_salts`, and `cs_db%num_cs`. The cs and salt readers downstream allocate their arrays from these counts.

The readers `cs_roadsalt_read` does not exist for cs. The cs readers `cs_uptake_read`, `cs_urban_read`, and `cs_reactions_read` only proceed when `cs_db%num_cs > 0`; the remaining readers test only for file existence.

All cs readers are called from `proc_read.f90` except `cs_cha_read`, which is called later from `main.f90`.

Files

constituents.cs

Master list of constituents.

- Reader: src/constit_db_read.f90
- Type: `cs_db` (`type cs_db_type`) in src/constituent_mass_module.f90.
- Format: title, then five pairs of `count` followed by `count` names, in order: pesticides, pathogens, metals, salts, other constituents. The metals block is read but is unused downstream (the matching loop is commented out in the source). The salt names are matched against the database in `salt_plants` by plant-parameter index, not by name.

cs_hru.ini

Initial constituent concentrations for HRU soils and plants.

- Reader: src/cs_hru_read.f90
- Type: `cs_soil_ini` (`type cs_soil_init_concentrations`) in `src/constituent_mass_module.f90`. Fields: `name`, `soil(2*num_cs)`, `plt(2*num_cs)` in ppm.
- Format: 4 header lines, then for each entry three lines: name, soil values, plant values.

cs_aqu.ini

Initial constituent concentrations for aquifers.

- Reader: src/cs_aqu_read.f90
- Type: `cs_aqu_ini` (`type cs_aqu_init_concentrations`) in `src/constituent_mass_module.f90`. Fields: `name`, `aqu(2*num_cs)` (dissolved plus sorbed).
- Format: 3 header lines, then one line per entry: `name`, `2*num_cs` values.

cs_channel.ini

Initial constituent concentrations for channels.

- Reader: src/cs_cha_read.f90
- Type: `cs_cha_ini` (`type cs_cha_init_concentrations`) in `src/constituent_mass_module.f90`. Fields: `name`, `conc(num_cs)` in g/m3.
- Format: 2 header lines, then one line per record: `name`, `num_cs` concentrations.

The same reader also processes the optional `cs_streamobs` file, which lists channels for which daily concentration and load output is written to `cs_streamobs_output`.

cs_irrigation

Constituent concentrations for irrigation water sourced from outside the watershed.

- Reader: src/cs_irr_read.f90
- Type: `cs_water_irr` (`type cs_irrigation_concentrations`) in `src/constituent_mass_module.f90`. Fields: `name`, `water(num_cs)` in ppm.
- Format: 2 header lines, then one line per source: `name`, `num_cs` concentrations.

cs_plants_boron

Plant boron tolerance parameters.

- Reader: src/cs_plant_read.f90
- Targets in src/cs_data_module.f90:

Field	Type	Units	Description
bor_tol_sim	int	flag	0 = off, 1 = simulate boron effect on plant growth
bor_stress_a(plantparm)	real	none	a parameter in the boron relative-yield equation
bor_stress_b(plantparm)	real	none	b parameter in the boron relative-yield equation

Format: title, blank, `bor_tol_sim`, four header lines, then one line per plant in plant database order with `name, a, b`.

cs_uptake

Specified daily constituent uptake (kg/ha) per plant community per constituent.

- Reader: src/cs_uptake_read.f90
- Targets in src/cs_module.f90: `cs_uptake_kg(plantparm, num_cs)`, `cs_uptake_on` (flag set to 1 when the file is present).
- Format: 3 header lines, then one line per plant: `name`, `num_cs` uptake values.

cs_reactions

Selenium and boron chemical reaction parameters for soils and aquifers, organised by reaction group and shale geology.

- Reader: src/cs_reactions_read.f90
- Type: `cs_rct_soil(sp_ob%hru)` and `cs_rct_aqu(sp_ob%aqu)` (`type constituent_rct`) in `src/cs_data_module.f90`. Also fills the module-level arrays `rct(num_rct, num_groups)` and `rct_shale(num_geol_shale, 3)`.
- Format: title, header, `num_rct num_groups`, `num_rct` lines of group values, header, `num_geol_shale`, `num_geol_shale` lines of three shale parameters, header, one line per HRU (`hru_id, group, shale_fractions...`), header, one line per aquifer (`aqu_id, group, shale_fractions...`). The eight `rct` rows hold (in order): inorganic-N to Se ratio, soil oxygen, aquifer oxygen, kd_seo4, kd_seo3, kd_boron, kseo4 in soil, kseo4 in aquifer. See the source for the full assignment table.

cs_urban

Constituent concentration in suspended solids from urban impervious areas (mg cs / kg sediment).

- Reader: src/cs_urban_read.f90
- Target: `cs_urban_conc(urban, num_cs)` in `src/cs_module.f90`.
- Format: 2 header lines, then one line per urban land-use type: `urb_type`, `num_cs` concentrations. The urban name must match an entry in `urban.urb`.

fertilizer.frt_cs

Per-fertilizer constituent loading (kg/ha).

- Reader: src/cs_fert_read.f90
- Type: `fert_cs(fertparm)` (`type fert_db_cs`) in `src/cs_module.f90`. Sets `fert_cs_flag = 1`.

#	Field	Units	Description
1	fertnm	-	fertilizer name
2	seo4	kg/ha	selenate loading
3	seo3	kg/ha	selenite loading
4	boron	kg/ha	boron loading

Format: title, header, then one line per fertilizer in the fertilizer database order.

Related

- salt module. The salt readers share `constituents.cs` for ion counts and use the same per-object pattern.
- fertilizer.frt. Base fertilizer database. `fertilizer.frt_cs` must have the same row count.
- urban.urb. Source of urban land-use names for `cs_urban`.
- atmodep.cli. Atmospheric deposition stations used by the cs deposition reader `cli_read_atmodep_cs`.